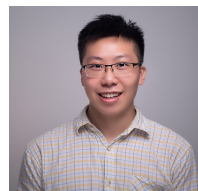


GUANGBIN CHEN

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EDUCATION

Huazhong University of Science and Technology, Computer Science and Technology, M.S.

Candidate

Sep 2019 – Sep 2025

First-Class Academic Scholarship, Outstanding Communist Youth League Member

Lanzhou University, Computer Science and Technology, B.S.

Sep 2015 – Jun 2019

Ranked 2/69 (top 3%), **National Scholarship**, Outstanding Graduate of Lanzhou University

COMPETITIONS & AWARDS

- **ACM-ICPC Regional Silver Medal** (2018 Asia Regional Jiaozuo Silver, 2018 Asia Regional Qingdao Silver)
- **Higher Education Press Mathematical Modeling Competition, Undergraduate National Second Prize**

INTERNSHIP EXPERIENCE

UDrive Innovation Intelligent Cockpit R&D C++ Shenzhen Aug 2025 – Present

- **Source Code Analysis:** Conducted in-depth studies of the occupant monitoring system (OMS) and driver monitoring system (DMS), focusing on facial and torso detection, facial image registration, and recognition algorithms.
- **Driver Hand Occupancy Detection Module:**
 - **Problem Context:** In autonomous driving scenarios, drivers may struggle to quickly take over when both hands stay off the steering wheel for extended periods. The system must continuously detect hand occupancy status and plan safe takeover strategies. This project, developed jointly with an OEM, aimed to deliver a functional initial release.
 - **Deep Learning Training, Inference, and Optimization:** Studied the full pipeline of lightweight hand and object detection models, from training to inference. Designed unit and integration tests, uncovering and fixing three technical defects.
 - **Module Integration and System Design:** Integrated the hand occupancy module into OMS, completing pre- and post-processing for object detection. Redesigned the NMSMerger module and related data structures for hand and object detection to satisfy new scenario requirements.
- **Automated Facial Registration/Recognition Testing:** Built automation scripts to batch process hundreds of video samples for facial registration and recognition, capturing failure cases for both registration and recognition.

Samsung China Android SystemUI Development Java, Kotlin Guangzhou Jun 2025 – Jul 2025

- **SystemUI Source Code Study:** Analyzed Android System UI source code with a focus on internal implementation of core components (Tiles, etc.).
- **Network Speed Indicator Design:** Investigated cross-version porting (Android 15 to 16) of the network speed indicator, covering dependencies on network state, toggle status, and screen on/off conditions, and produced UML sequence diagrams.

ALGORITHM EXPERIENCE

Swin Transformer for Cerebral Vascular Segmentation, Skeletonization, and Network Analysis

Jan 2024 – May 2025

- **Background:** Cross-disciplinary collaboration on brain vasculature research. Tsinghua University (Academician Zhuo Zhuang) led physical experiments; Huazhong University of Science and Technology (Academician Qingming Luo) led imaging with ex-vivo advantages (complete, fine-grained, spatially locatable data). Responsible for full-brain 3D vascular segmentation, skeletonization, and graph analysis. The task demands large receptive fields to capture injury signatures and must mitigate vessel breaks and noise.

- **Methods:** Evaluated 19 supervised and zero-shot methods; a Swin Transformer with multi-level feature fusion, self-attention, and sliding windows performed best. Adapted Patch Embedding/Merging, Window Partition/Attention from 2D to 3D, added SENet channel attention to improve completeness, stacked $5 \times 1 \times 1$ and $3 \times 3 \times 3$ convolutions to enhance z-axis and small-scale connectivity, and incorporated lightweight denoising plus checkerboard artifact suppression. Delivered full segmentation, skeletonization, and visualization for 15 healthy or injured datasets every 2–3 days, enabling hemodynamic simulations and pre-clinical studies.
- **Tech Stack:** PyTorch, NumPy, SciPy, SimpleITK/nibabel/libtiff (medical imaging), NetworkX (skeletonization and graph analysis), DataParallel, TensorBoard. Training on four NVIDIA A100 80GB GPUs; inference across fifteen NVIDIA RTX 5000 32GB servers. Portions are open-sourced on GitHub.

DEVELOPMENT EXPERIENCE

- **Mobile System Development:**
 - Studied Android System UI source with emphasis on porting the network speed indicator from Android 15 to 16, gaining architectural and state management knowledge.
 - Extensive iOS ecosystem usage (iPhone, iPad, Mac, Watch, AirPods); researched iOS 16 features such as Liquid Glass and Snipaste (**open-source**).
 - Accumulated hundreds of hours interacting with Apple Support to resolve issues and propose improvements, building deep insights into the iOS ecosystem.
- **Engineering Tooling:**
 - Managed 14 workflows across 15 servers for deep learning projects via automation scripts, achieving efficient distributed scheduling and monitoring.
 - Web literature translation (**open-source**): Built Power Automate + Tampermonkey scripts for bulk translation, including auto-scrolling, navigation, and math-aware translation bypass.
 - GitHub Projects to PDF (**open-source 1** / **open-source 2**): Automated conversion of GitHub resources into PDFs, handling crawler limits, image loading, Markdown/HTML formatting, and auto-click flows.
 - Resume LaTeX template (**open-source**): Dynamically optimizes margins and paragraph widths, supports multi-version resumes assembled via conditional logic, and auto-generates PDFs per target role.
- **Full-Stack Development:**
 - Article reading web app with user interactions (**open-source**) (Front-end: Vue.js, HTML/CSS, JavaScript, HTTP; Back-end: Python, Django.db): CRUD for users/articles, reactions (like/favorite/comment), and author dashboards.
 - Movie recommendation system (**open-source**) (C, WinAPI for windowing, message handling, controls, drawing, file I/O): Rich GUI with scenario-based recommendations.
 - Metro online ticketing system (**open-source**) (Java, GUI, Socket, Thread, MySQL, algorithms): Multi-layer maps, zooming, multiple interaction modes, custom distance calculations.
- **Tooling Expertise:**
 - **Development & Debugging:** Proficient with Git, CMake, gdb, perf, valgrind, etc.
 - **Data Management:** Responsible for petabyte-scale lab cluster backups using HDDs, tapes, optical media, etc.

GRAPH THEORY (ROUTING) EXPERIENCE

- **Algorithms Mastered/Familiar:**
 - Fundamentals: DFS/BFS, Dijkstra, Bellman-Ford, SPFA, Kruskal, Prim, Topological Sort, Tarjan, Max-Flow, Hamiltonian Graphs, A*
 - Related Algorithms: LCA, heavy-light decomposition, tree decomposition, tree diameter, segment trees, bitmask DP, dynamic programming, meet-in-the-middle
 - Advanced: TSP, VRP, reinforcement learning, approximation algorithms
- **Blogging:** Authored 500+ posts covering graph algorithms and research work.
- **Problem Setting (open-source):** Created and verified DFS/BFS/multi-source BFS/Dijkstra/Hamiltonian graph/flow problems.
- **Applications:** Designed a mobile robot trash bin with dual rollers for waste collection.